**10** Two of the isotopes of hydrogen are hydrogen-2 ( $^{2}_{1}H$ ) and hydrogen-3 ( $^{3}_{1}H$ ).

**(a) (i)** State **one** similarity in the composition of their nuclei.

..... [1]

**(ii)** Describe how a nucleus of hydrogen-3 differs from a nucleus of hydrogen-2.

.....

..... [2]

**(b)** In a nuclear fusion reactor, a nucleus of hydrogen-2 fuses with a nucleus of hydrogen-3 at an extremely high temperature. This fusion reaction produces an isotope of element X and releases a neutron.

**(i)** Explain why an extremely high temperature is needed when forcing these two nuclei together.

.....

.....

.....

..... [3]

**(ii)** Using nuclide notation, complete the equation for this reaction.

$$^{2}_{1}H + {}^{3}_{1}H \longrightarrow$$

[2]

[Total: 8]

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